

Calculating Artifact Roll Outcomes

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This document provides an overview of the approach taken to calculate all possible outcomes for an artifact in Genshin Impact.

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1 Sub-Stat Probability

Unique combinations are computed and the probability of each corresponding permutation is calculated based on the stat weights.

The formula for the likelihood of a permutation, where w_i is the weight of sub-stat i , is:

$$\left(\frac{w_1}{Total}\right) \left(\frac{w_2}{Total - w_1}\right) \left(\frac{w_3}{Total - w_1 - w_2}\right) \cdots$$

For a given unique combination, all of these probabilities are added together for all permutations that correspond to that unique combination.

The sub-stat probability is then the sum of probabilities of combinations that are valid given the inputted constraints,

$$SubstatProbability = \sum_{valid} p$$

2 Roll Probability

The probability for a given sub-stat outcome is:

$$p \left(\frac{1}{InitialValueOutcomes}\right) \left(\frac{1}{RollOutcomes}\right) \left(\frac{1}{RollValueOutcomes}\right)$$

Where:

- p is the probability of the stat combination.
- $InitialValueOutcomes$ is the number of possible roll outcomes.
- $InitialValueOutcomes$ the number of value outcomes for the initial states.
- $RollValueOutcomes$ the number of value outcomes for the rolled stats.

Since the sub-stat probability is separately defined above, we have:

$$RollProbability \times SubstatProbability =$$

$$p \left(\frac{1}{InitialValueOutcomes}\right) \left(\frac{1}{RollOutcomes}\right) \left(\frac{1}{RollValueOutcomes}\right)$$

$$RollProbability \times \sum_{valid} p =$$

$$p \left(\frac{1}{InitialValueOutcomes}\right) \left(\frac{1}{RollOutcomes}\right) \left(\frac{1}{RollValueOutcomes}\right)$$

$$RollProbability =$$

$$\left(\frac{p}{\sum_{valid} p}\right) \left(\frac{1}{InitialValueOutcomes}\right) \left(\frac{1}{RollOutcomes}\right) \left(\frac{1}{RollValueOutcomes}\right)$$

As there are 4 stats to choose from each time, and since there are 4 roll values a roll can have (70%, 80%, 90%, and 100%), all these are exponents with a base of 4. Let n be the number of rolls.

- $InitialValueOutcomes = 4^4$ - There are 4 initial substats
- $RollOutcomes = 4^n$ - There are n sub-stats to roll
- $RollValueOutcomes = 4^n$ - n sub-stats where rolled, and need a value assigned

Thus, the total number of possibilities is:

$$(4^n)^2(4^4)$$

2.1 Roll Permutations

To preserve the base of 4, each guaranteed roll is counted as $\frac{4}{\#ofguaranteedrolls} = \frac{4}{2} = 2$.

To reduce computation, the computed 4^n roll permutations are combined into combinations, giving each effective outcome a value *RollPermutations*.

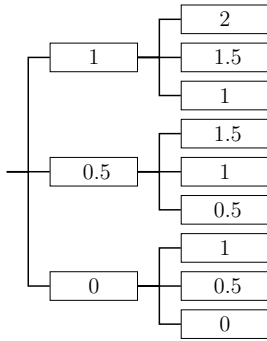
2.2 Roll Values

For each of these, the number of good outcomes is found based on the possible roll values.

For n rolls, there are $4^{a+1}4^{b+1}4^{c+1}4^{d+1}$ possible roll value combinations where $a+b+c+d = n$.

A method computes the roll value distribution of $x = a + 1, \dots, d + 1$. It iterates x times, considering the 4 possible roll values that can extend the current computed values each time. To avoid computing a full 4^x outcomes, after each stage, paths with the same value are combined.

For example, assume the roll values are 0, 0.5, and 1. After the second iteration, we have:



Giving us $3^2 = 9$ outcomes to bring to the next iteration.

But instead, they can be grouped as so: 2 (1x), 1.5 (2x), 1 (3x), 0.5 (2x), 0 (1x). This leaves only 5 outcomes to bring to the next iteration. This is repeated for all of the x iterations.

The general method to compute the number of outcomes above the goal also uses this method, but for the 4 stat iterations.

This gives each outcome $RollValuePossibilities_i$ for $i = 1, \dots, 4$.

2.3 Total Probability

The probability associated with a single outcome is now:

$$\frac{p \times \text{RollPermutations} \times \prod_{j=1}^4 \text{RollValuePossibilities}_j}{\sum_{\text{valid}} p \times \text{RollOutcomes} \times \text{InitialValueOutcomes} \times \text{RollValueOutcomes}}$$

So, the total probability is:

TotalProbability

$$\begin{aligned} &= \sum_{i=1}^{|\text{combos}|} \left(\frac{p \times \text{RollPermutations} \times \prod_{j=1}^4 \text{RollValuePossibilities}_j}{\sum_{\text{valid}} p \times \text{RollOutcomes} \times \text{InitialValueOutcomes} \times \text{RollValueOutcomes}} \right) \\ &= \frac{\sum_{i=1}^{|\text{combos}|} p_i \times \text{RollPermutations}_i \times \prod_{j=1}^4 \text{RollValuePossibilities}_{i,j}}{\sum_{\text{valid}} p \times \text{RollOutcomes} \times \text{InitialValueOutcomes} \times \text{RollValueOutcomes}} \\ &= \frac{\sum_{i=1}^{|\text{combos}|} p_i \times \text{RollPermutations}_i \times \prod_{j=1}^4 \text{RollValuePossibilities}_{i,j}}{\sum_{\text{valid}} p \times (4^n)^2 (4^4)} \end{aligned}$$

2.4 Generating Outcomes

An overview of the code used to generate all outcomes is thus:

Sum = 0

```
foreach ( $s_1, s_2, s_3, s_4 \rightarrow p$ ) of GenerateCombos do
  foreach ( $r_1, r_2, r_3, r_4 \rightarrow \text{RollPermutations}$ ) of RollCountsDist do
    States = Map( $0 \rightarrow 1$ )
    for  $i = 1 \dots 4$  do
      NextStates = Map()
      foreach ( $\text{State} \rightarrow \text{Ways}$ ) of States do
        foreach ( $\text{statSum}_i \rightarrow \text{RollValuePossibilities}_i$ ) of ValueOfRollsDist( $r_i + 1$ ) do
          NextState =  $\text{State} + \text{statSum}_i \times \text{weight}_i$ 
          NextStates[NextState] +=  $\text{Ways} \times \text{RollValuePossibilities}_i$ 
        end
      end
      States = NextStates
    end
    foreach ( $\text{State} \rightarrow \text{Ways}$ ) of States do
      if State is within goal then
        Sum +=  $p \times \text{RollPermutations} \times \text{Ways}$ 
      end
    end
  end
end
return  $\text{Sum} \div (\sum_{\text{valid}} p \times (4^n)^2 (4^4))$ 
```

Note that at the end, *Ways* is just the product of the *RollValuePossibilities_i*, $i = 1, \dots, 4$, for the outcome.

The distribution viewer provides a way to visualize *RollCountsDist* and *ValueOfRollsDist*.

2.5 Varying Initial Stat Count

Since artifacts have a chance to have 4 or 5 rolls (corresponding to 3 or 4 initially enabled stats), the actual probability is,

$$TotalProbability_{n=4} \times (1 - P(n = 5)) + TotalProbability_{n=5} \times P(n = 5)$$

So, the above pseudo-code is called twice for each calculation.